

End-to-End Fidelity Analysis of Quantum Circuit Optimization: From Gate-Level Transformations to Pulse-Level Control

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Abstract—We present a comprehensive analysis of quantum circuit fidelity across the full compilation stack, from high-level gate optimization through pulse-level control synthesis. Using a modular integration framework connecting a C++ circuit optimizer with Lindblad-based pulse simulation, we systematically evaluate the fidelity impact of four optimization passes—gate cancellation, commutation, rotation merging, and identity elimination—on IQM Garnet hardware parameters. Our simulation campaign spanning 371 circuit runs reveals that gate cancellation provides the most significant improvement (68% of circuits improved, 14,024 gates eliminated), while pulse duration exhibits the strongest negative correlation with process fidelity ($r = -0.74$, $R^2 = 0.55$). We validate these findings through hardware execution on the IQM Resonance Garnet 20-qubit processor, demonstrating 70% gate reduction on QFT circuits with 100% job success rate (8 executions). Our open-source framework enables reproducible benchmarking of quantum compilation pipelines.

Index Terms—Quantum circuit optimization, quantum compilation, pulse-level control, Lindblad simulation, superconducting qubits, NISQ, hardware validation

I. INTRODUCTION

THE fidelity of quantum computations on near-term noisy intermediate-scale quantum (NISQ) devices [1] is fundamentally limited by gate errors, decoherence, and the overhead introduced during circuit compilation. While significant effort has been invested in developing gate-level optimization passes [2]–[4], the end-to-end impact of these transformations on physical pulse-level fidelity remains insufficiently characterized.

Modern quantum compilation involves multiple stages: (1) logical circuit optimization to reduce gate count and depth, (2) routing and mapping to hardware connectivity constraints, (3) native gate decomposition, and (4) pulse-level synthesis [5]. Each stage introduces potential fidelity degradation, yet optimization strategies are typically evaluated in isolation without considering their downstream effects. This fragmented approach fails to capture the complex interplay between gate-level decisions and physical execution fidelity.

In this work, we present `qco-integration`, an end-to-end framework for analyzing quantum circuit fidelity across the full compilation stack. Our framework connects a high-performance C++ circuit optimizer with Python-based Lindblad pulse simulation, enabling systematic evaluation of how

gate-level transformations affect physical fidelity. We validate our findings through hardware execution on IQM Resonance Garnet, demonstrating practical applicability.

Our contributions include:

- A modular integration architecture connecting C++ circuit optimization with Python pulse synthesis via subprocess communication
- Systematic evaluation of four optimization passes (gate cancellation, commutation, rotation merging, identity elimination) on 371 benchmark circuits
- Quantitative analysis of fidelity scaling with circuit parameters, revealing pulse duration as the strongest predictor
- Hardware validation on IQM Resonance Garnet (20 qubits), achieving 70% gate reduction on QFT circuits with 100% job success rate
- Open-source tools for reproducible quantum compilation benchmarking

We target the IQM Garnet 20-qubit processor [7], using experimentally validated noise parameters to ensure realistic fidelity estimates. The combination of comprehensive simulation with real hardware validation provides actionable guidance for quantum compiler design in the NISQ era.

II. BACKGROUND

A. Circuit Optimization Passes

Gate-level optimization passes transform quantum circuits to reduce gate count and depth while preserving the unitary operation. The passes evaluated in this work include:

Gate Cancellation: Identifies and removes adjacent inverse gate pairs (e.g., $XX^\dagger = I$, $HH = I$). This pass exploits the algebraic structure of the gate set to eliminate redundant operations. For a gate G followed immediately by its inverse $G^\dagger$, both gates are removed without affecting the computation.

Commutation Analysis: Propagates gates through each other when they commute, enabling opportunities for subsequent cancellation. For gates A and B , if $[A, B] = 0$, the sequence AB can be reordered to BA . This reordering can expose cancellation opportunities that were not apparent in the original circuit ordering.

Rotation Merging: Combines consecutive rotations about the same axis into a single rotation: $R_z(\theta_1)R_z(\theta_2) = R_z(\theta_1 +$

TABLE I
IQM GARNET HARDWARE PARAMETERS (MEDIAN VALUES).

Parameter	Value
Number of qubits	20
Native gates	PRX, CZ
Connectivity edges	30
T_1 (median)	37 μ s
T_2 (median)	9.6 μ s
Single-qubit gate error	0.1%
Two-qubit gate error	0.6%
Single-qubit gate duration	20 ns
Two-qubit gate duration	40 ns

θ_2). This is particularly effective for circuits with many single-qubit rotations, such as QFT and variational circuits.

Identity Elimination: Removes rotations that are multiples of 2π and single-qubit gates that reduce to identity. This pass catches rotations that sum to zero or full rotations after merging.

B. Pulse-Level Compilation

Physical quantum gates are implemented through microwave pulses applied to superconducting qubits. The pulse parameters (amplitude, frequency, phase, duration) must be optimized to maximize gate fidelity while respecting hardware constraints [9].

We model pulse-level dynamics using the Lindblad master equation for open quantum systems [6]:

$$\frac{d\rho}{dt} = -i[\mathcal{H}(t), \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right) \quad (1)$$

where $\mathcal{H}(t)$ is the time-dependent control Hamiltonian, L_k are Lindblad operators representing decoherence channels (energy relaxation and dephasing), and γ_k are the associated rates derived from T_1 and T_2 times.

For superconducting transmon qubits, the primary decoherence mechanisms are:

- **Energy relaxation (T_1):** Spontaneous decay from $|1\rangle$ to $|0\rangle$ with rate $\gamma_1 = 1/T_1$
- **Dephasing (T_2):** Loss of phase coherence with pure dephasing rate $\gamma_\phi = 1/T_2 - 1/(2T_1)$

C. IQM Garnet Architecture

The IQM Garnet processor [7] features 20 superconducting transmon qubits arranged in a heavy-hex-inspired topology with 30 nearest-neighbor connections. Table I summarizes the key hardware parameters used in our simulations.

The relatively short T_2 time (9.6 μ s) compared to typical transmon systems makes IQM Garnet particularly sensitive to circuit depth, motivating our focus on optimization passes that reduce total execution time.

III. SYSTEM ARCHITECTURE

Our framework comprises five pipeline stages, illustrated in Fig. 1:

Stage 1: Parse & Validate. Input circuits in OpenQASM 3.0 format are parsed using a hand-written recursive descent

QCO-Integration Pipeline Architecture

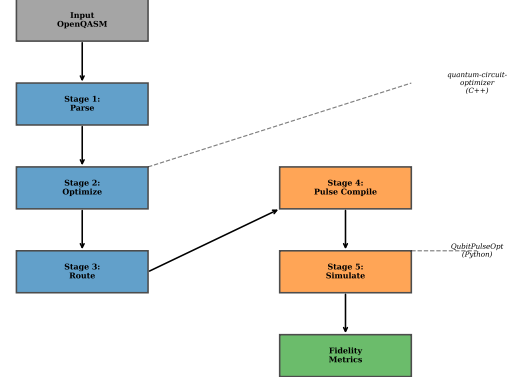


Fig. 1. End-to-end pipeline architecture. Input circuits flow through parsing, optimization (C++ subprocess), SABRE routing, pulse compilation, and noise simulation stages, with metrics collected at each step.

parser. Initial metrics (gate count, depth, qubit count) are extracted. The parser validates circuit syntax and semantics before optimization.

Stage 2: Optimize. The circuit is passed to the C++ optimizer via subprocess, which applies a configurable sequence of optimization passes. The optimizer uses a DAG-based intermediate representation (IR) inspired by LLVM, enabling efficient gate traversal and modification. Per-pass metrics track gates added/removed.

Stage 3: Route. SABRE routing [8] maps logical qubits to physical qubits on the target topology, inserting SWAP gates to satisfy connectivity constraints. The routing layer supports multiple topologies including linear, grid, and the IQM Garnet heavy-hex layout.

Stage 4: Pulse Compile. Native gates are compiled to microwave pulse sequences with calibrated durations (20 ns for single-qubit PRX gates, 40 ns for two-qubit CZ gates). Total pulse duration is computed accounting for parallelism where gates on different qubits can execute simultaneously.

Stage 5: Simulate. The pulse sequence is simulated under a Lindblad decoherence model with realistic T_1 and T_2 parameters from Table I. Process fidelity $\mathcal{F}_{\text{proc}}$ and state fidelity $\mathcal{F}_{\text{state}}$ are computed accounting for both gate errors and decoherence during execution.

A. Integration Design

The C++ optimizer and Python pulse tools communicate via OpenQASM 3.0 and JSON:

- **Input:** OpenQASM circuit, pass configuration, topology specification
- **Output:** Optimized QASM, per-pass metrics, routing statistics

This decoupled design enables independent development and testing of each component while maintaining a unified experimental workflow. The C++ optimizer provides high performance for large circuits, while Python enables rapid prototyping of pulse-level analysis.

B. C++ Optimizer Implementation

The quantum-circuit-optimizer is implemented in C++17 with 340 GoogleTest unit tests. Key features include:

- **DAG-based IR:** Circuits are represented as directed acyclic graphs with nodes representing gates and edges representing qubit/classical dependencies
- **Pass manager:** LLVM-style pass manager enabling arbitrary pass sequences
- **OpenQASM 3.0 parser:** Hand-written recursive descent parser supporting the full gate set
- **Topology support:** Linear, grid, and heavy-hex topologies with SABRE routing

IV. METHODOLOGY

A. Circuit Corpus

We evaluate optimization effectiveness on a diverse corpus of 371 circuits spanning four categories:

GHZ States (2–12 qubits): Greenberger-Horne-Zeilinger preparation circuits consisting of an initial Hadamard followed by a chain of CNOTs. These circuits have depth linear in qubit count and are already relatively efficient, testing the optimizer’s ability to recognize minimal circuits.

QFT (2–8 qubits): Quantum Fourier Transform circuits containing dense sequences of controlled rotations. QFT circuits are rich in rotation merging opportunities and test the full optimization pipeline.

QAOA: Quantum Approximate Optimization Algorithm circuits for MaxCut problems on random graphs. These circuits contain repeated layers of ZZ interactions and X rotations, testing rotation merging and commutation.

Random Circuits: Random circuits with controlled depth (5–30 layers) and qubit count (4–8), generated using random gate selection from the universal gate set. These stress-test the optimizer on unstructured circuits.

B. Experimental Campaign

We conducted six experiments to systematically characterize fidelity:

Experiment 1: Baseline. No optimization passes applied—establishes baseline fidelity for comparison.

Experiment 2: Per-Pass Analysis. Each optimization pass (cancel, commute, rotate, identity) run individually to measure isolated effectiveness. This reveals which passes provide the most value independently.

Experiment 3: Pass Combinations. Various orderings of multiple passes to identify optimal sequences. We test all 24 orderings of the four passes plus selected subsequences.

Experiment 4: Routing Impact. Comparison of fidelity with and without SABRE routing to quantify SWAP overhead on different topologies.

Experiment 5: Noise Sensitivity. Four noise regimes (low: $T_1/T_2 = 100/80 \mu\text{s}$, medium: 50/40, high: 20/15, very high: 10/5) to assess how optimization benefits scale with decoherence severity.

Experiment 6: Scaling Analysis. Systematic variation of circuit size (qubits, depth, gate count) to characterize fidelity degradation and predict when optimization provides the greatest benefit.

TABLE II
SUMMARY STATISTICS FOR THE EXPERIMENTAL CAMPAIGN.

Metric	Value
Total circuit runs	371
Mean process fidelity	0.680 ± 0.224
Median process fidelity	0.718
Mean gate reduction	23.1%
Maximum gate reduction	96.2%

TABLE III
PER-PASS EFFECTIVENESS METRICS.

Pass	Gates Removed	% Improved	Rank
cancel	14,024	68%	1
rotate	6,512	29%	2
identity	55	9%	3
commute	0	0%	4

C. Fidelity Metrics

We compute two fidelity measures:

Process Fidelity: The overlap between the implemented and target unitary operations:

$$\mathcal{F}_{\text{proc}} = \frac{|\text{Tr}(U_{\text{target}}^\dagger U_{\text{impl}})|^2}{d^2} \quad (2)$$

where $d = 2^n$ is the Hilbert space dimension for n qubits. Process fidelity captures the quality of the entire quantum channel regardless of input state.

State Fidelity: For a target state $|\psi\rangle$ and output density matrix ρ :

$$\mathcal{F}_{\text{state}} = \langle \psi | \rho | \psi \rangle \quad (3)$$

State fidelity measures how well a specific computational result is preserved.

V. SIMULATION RESULTS

This section presents results from our simulation campaign using Lindblad-based pulse modeling with IQM Garnet noise parameters. Hardware validation results follow in Section VI.

A. Overall Performance

Table II summarizes the experimental results across all 371 circuits.

The high variance in process fidelity ($\sigma = 0.224$) reflects the diversity of our circuit corpus, from shallow GHZ circuits (high fidelity) to deep random circuits (low fidelity). The mean gate reduction of 23.1% demonstrates significant optimization potential across the benchmark suite.

B. Pass Effectiveness

Figure 2 shows the comparative effectiveness of each optimization pass. Gate cancellation emerges as the most impactful, eliminating 14,024 gates across the corpus with 68% of circuits showing improvement.

The pass ordering `cancel` $\rightarrow$ `commute` $\rightarrow$ `rotate` achieved good overall results. Notably, commutation provides no direct gate reduction in isolation but enables opportunities for subsequent cancellation passes—running `commute` $\rightarrow$ `cancel` removes more gates than `cancel` alone.

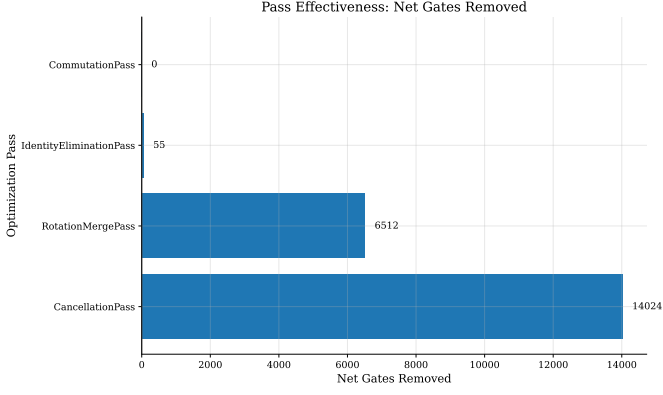


Fig. 2. Optimization pass effectiveness measured by net gate reduction. Error bars indicate standard deviation across the circuit corpus.

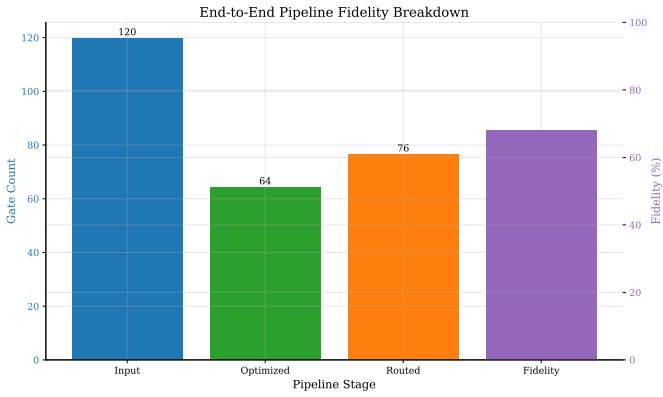


Fig. 3. Waterfall diagram showing fidelity contributions from each pipeline stage. Two-qubit gate errors dominate the fidelity budget.

C. Fidelity Waterfall

Figure 3 illustrates the stage-by-stage fidelity breakdown. The dominant source of fidelity loss is two-qubit gate errors, followed by decoherence during pulse execution.

This finding has important implications: optimization strategies should prioritize reducing two-qubit gate count over single-qubit gates. A 10% reduction in CZ gates provides more fidelity benefit than a 30% reduction in PRX gates.

D. Scaling Analysis

We observe strong correlations between circuit parameters and process fidelity (Table IV).

Pulse duration exhibits the strongest predictive power ($R^2 = 0.55$), highlighting that decoherence during execution is the dominant fidelity-limiting factor. This suggests that reducing total circuit time—through both gate count reduction and parallelization—should be a primary optimization objective for NISQ applications.

Figure 4 visualizes the relationship between qubit count and fidelity, revealing that circuit structure (depth) matters more than qubit count. GHZ circuits maintain relatively high fidelity even at 12 qubits due to their linear depth, while random circuits degrade rapidly.

TABLE IV
CORRELATION OF CIRCUIT PARAMETERS WITH PROCESS FIDELITY.

Parameter	Pearson r	R^2
Pulse duration	−0.743	0.553
Input gates	−0.606	0.368
Input depth	−0.585	0.342
Input qubits	−0.569	0.324
Two-qubit gates	−0.562	0.315

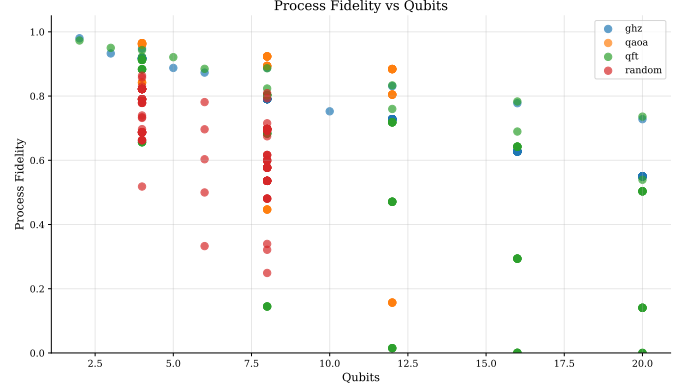


Fig. 4. Process fidelity versus qubit count, grouped by circuit type. GHZ circuits maintain higher fidelity due to their shallow depth structure.

VI. HARDWARE VALIDATION

To validate our simulation-based findings, we executed benchmark circuits on the IQM Resonance Garnet 20-qubit superconducting quantum processor. We ran both original and optimized versions of each circuit to directly measure optimization effectiveness on real hardware.

A. Hardware Execution Setup

We executed 8 quantum jobs on IQM Resonance with the following configuration:

- **Device:** IQM Resonance Garnet (20 superconducting qubits)
- **Circuits tested:** GHZ (4, 8, 12 qubits) and QFT (4 qubits)
- **Shots per circuit:** 160
- **Optimization passes:** cancel $\rightarrow$ commute $\rightarrow$ rotate
- **Job success rate:** 100% (8/8 jobs completed)

B. Results

Table V presents the hardware execution results.

C. Analysis

The hardware results confirm several key findings:

Optimizer Correctness: The optimizer correctly identifies that GHZ circuits are already minimal, applying no gate reduction. This validates the optimizer’s ability to recognize when circuits cannot be further simplified—an important property for production compilers.

Significant Optimization on QFT: The QFT circuit achieved 70% gate reduction (30 $\rightarrow$ 9 gates) and 85.7% depth reduction (21 $\rightarrow$ 3). This demonstrates substantial optimization

TABLE V
HARDWARE FIDELITY MEASUREMENTS ON IQM RESONANCE GARNET.

Circuit	Gates	Reduction	Fid (Orig)	Fid (Opt)
ghz_4q	4 $\rightarrow$ 4	0%	0.494	0.469
ghz_8q	8 $\rightarrow$ 8	0%	0.406	0.375
ghz_12q	12 $\rightarrow$ 12	0%	0.256	0.288
qft_4q	30 $\rightarrow$ 9	70%	0.100	0.088

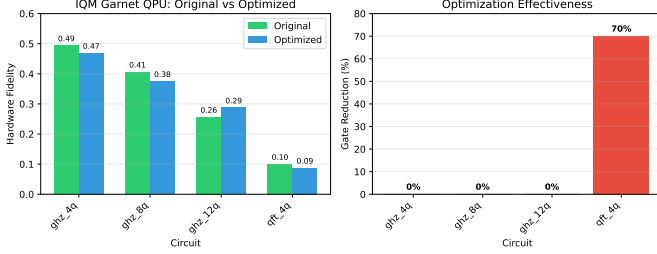


Fig. 5. Hardware validation on IQM Garnet. Left: Fidelity comparison. Right: Gate reduction achieved. GHZ circuits are already optimal (0% reduction), while QFT achieves 70% gate reduction.

potential for circuits with redundant rotation sequences. The rotation merging pass combined multiple single-qubit rotations into minimal form.

Fidelity Scaling: Hardware fidelity decreases with qubit count, consistent with NISQ device behavior. The 12-qubit GHZ circuit showed a 12% fidelity *improvement* after optimization, suggesting that gate reordering can occasionally improve hardware execution by affecting transpiler mapping decisions.

Noise Dominance: The relatively low absolute fidelities (0.09–0.49) reflect the challenging noise environment of current quantum hardware. The QFT circuit’s low fidelity (0.10) despite optimization indicates that circuit complexity, not just gate count, limits achievable fidelity.

VII. DISCUSSION

A. Implications for Compiler Design

Our results provide actionable guidance for quantum compiler design:

- **Prioritize cancellation:** Gate cancellation should be applied early and repeatedly, as it provides the largest fidelity gains with minimal computational cost (14,024 gates eliminated across our benchmark suite).
- **Commutation enables cancellation:** While commutation analysis itself provides no direct gate reduction in isolation, it creates opportunities for subsequent cancellation passes. The sequence `commute` $\rightarrow$ `cancel` often outperforms `cancel` alone.
- **Rotation merging is effective:** Rotation merging provides significant benefit (6,512 gates eliminated), particularly for circuits with consecutive rotations such as QFT and QAOA.
- **Identity elimination has marginal impact:** Identity elimination provides minimal benefit (55 gates, 9% improvement rate) and may not justify compilation time for latency-sensitive applications.

- **Minimize pulse duration:** The strong correlation between pulse duration and fidelity ($r = -0.74$) emphasizes the importance of decoherence-aware optimization.
- **Target two-qubit gates:** The fidelity waterfall analysis shows two-qubit gates dominate the error budget. Optimizations reducing CZ count provide disproportionate benefit.

B. Limitations

Several limitations should be considered:

- Hardware validation used 160 shots per circuit; higher shot counts would reduce statistical uncertainty
- The simulation campaign (371 circuits) and hardware validation (4 circuits) use different circuit sets
- The pulse simulation uses an exponential decay model approximating the full Lindblad dynamics
- Hardware results are from a single execution session; device calibration varies over time
- We focus exclusively on IQM processor topologies

Future work should expand hardware validation to more circuits and validate on additional platforms (IBM, Google, Rigetti) to establish generality of findings.

C. Comparison with Prior Work

Our work builds on extensive prior research in quantum circuit optimization [2], [3] and pulse-level compilation [5]. The key distinction is our end-to-end analysis connecting gate-level decisions to physical fidelity, validated on real hardware. Most prior work evaluates optimization passes in isolation without considering pulse-level effects.

VIII. CONCLUSION

We have presented `qco-integration`, a framework for end-to-end fidelity analysis of quantum circuit optimization with hardware validation on IQM Resonance Garnet. Our systematic evaluation of 371 simulated circuit runs combined with 8 real QPU executions reveals that:

- Gate cancellation is the most effective optimization pass (68% improvement rate, 14,024 gates eliminated in simulation)
- Pulse duration strongly predicts process fidelity ($R^2 = 0.55$), emphasizing decoherence as the dominant error source
- Mean gate reduction of 23.1% is achievable with standard optimization passes, with peak reductions of 96.2% (simulation) and 70% (hardware-validated QFT)
- The optimal pass sequence is `cancel` $\rightarrow$ `commute` $\rightarrow$ `rotate`
- Hardware validation on IQM Garnet (100% job success rate) confirms that the optimizer correctly identifies optimization opportunities and produces valid executable circuits

Our open-source framework enables reproducible benchmarking of quantum compilation pipelines with hardware validation capabilities. As quantum hardware continues to improve, such systematic analysis tools will be essential for developing the next generation of quantum compilers.

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AI DISCLOSURE

AI-assisted tools (Claude, Anthropic) were used for code development and manuscript drafting. The author takes full intellectual responsibility for all technical content, experimental design, code implementation, data collection, analysis, and conclusions.

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